

ERNESTINE TINGWEI

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Summary

- DevOps Engineer with extensive experience in managing complex infrastructure and deploying technologies to streamline software deployment. Skilled in onboarding CI/CD pipeline projects using tools like Terraform, Ansible, and Kubernetes. Successfully automated processes to reduce response times and improve infrastructure efficiency. Eager to leverage expertise in cloud solutions and DevOps practices to enhance operational performance.

Skills

- **AWS Network security**, and system security., Virtual and Cloud Technologies., Cloud security tools such as Aquasec, Inspector, and Dome9., Vulnerability Management.
- **Monitoring & Event Management:** AWS Cloud Watch (Events & Logs), AWS SNS, AWS S3., SIEM.
- **AWS Security:** AWS Security Hub, AWS Guard Duty, AWS Shield, AWS Firewall Manager, AWS Inspector, etc.
- **Cloud platforms:** AWS, Azure., AWS Services: EC2, S3, Lambda, RDS, IAM, CloudFormation, and more.
Azure Services: VMs, Blob Storage, Functions, SQL Database, RBAC, and more., Cloud security best practices. **Programing and Scripting languages:** python, Shell scripting, Bash, PowerShell., Communication,
- Containerization and Orchestration: Docker, Kubernetes, Container, Helm., Infrastructure as code: Terraform, AWS CloudFormation, Ansible, Puppet, Chef, Pulumi.
- **Databases:** AWS RDS, Azure SQL Database, MySQL, PostgreSQL, NoSQL databases., Networking: VPC, Vets, security groups, load balancers.
- **CI/CD Tools:** Jenkins, GitLab CI/CD, GitHub Actions, Azure DevOps.,
- **Serverless Computing:** AWS Lambda, Azure Functions., Version Control: Git, Gitlab, GitHub, Bitbucket.,
- Monitoring and logging: Prometheus, Grafana, ELK Stack, Splunk, Azure Log Analytics, AWS CloudWatch., Security: IAM Policies, Active Directory (Azure AD), security scanning tools (e.g., SonarQube, Nessus, Burp Suite, OWASP ZAP), network security groups., Performance Management, Continuous Integration, Project Planning

Work Experience

Nelnet Inc.

Cloud and DevOps Engineer

Apr 2024 - Present

- Automated routine tasks using Python and Bash scripting languages and developed custom tools, enhancing team productivity by streamlining processes
- Automated various operational tasks, including patching Windows and Linux servers across multiple environments, and designed workflows for domain controllers and SQL servers, ensuring minimal downtime and consistency
- Applied Agile methodologies, System Development Lifecycles (SDLCs), and DevOps best practices to improve project delivery timelines and enhance team collaboration
- Managed and maintained AWS resources using Terraform and CloudFormation, ensuring consistent and repeatable infrastructure deployment, which improved system reliability
- Implemented monitoring and alerting systems using Prometheus, Grafana, and Cribl to proactively identify performance bottlenecks and issues, enabling faster incident response and resolution
- Designed and implemented AWS database solutions, including Amazon RDS, Amazon Aurora, Amazon DynamoDB, and Amazon Redshift, to effectively meet business requirements
- Automated database provisioning and maintenance tasks using AWS CloudFormation templates, reducing manual intervention and enhancing infrastructure as code practices
- Reduce the average response time by 80% through the automation of manual processes using 65 Ansible modules.
- Orchestrated containerized applications using Docker and Kubernetes, ensuring scalability and high availability of microservices
- Implemented Identity and Access Management (IAM) policies and role-based access control to secure cloud environments on AWS
- Documented AWS Glue job configurations, transformations, and best practices for team reference, promoting knowledge sharing and consistency, which improved team efficiency
- Designed and developed automated solutions using AWS CloudFormation and Terraform, reducing provisioning time and cost by 35%
- Designed Jenkins open-source automation server to install code-built artifacts with code deployment and formulated several functioning CI/CD pipelines, improving deployment efficiency
- Developed custom security tools and scripts in bash to automate security tasks, enhancing the security posture of the organization and reducing manual effort
- Collaborated with cross-functional teams to design and implement SRE best practices including Service Level Objectives

(SLOs), Service Level Indicators (SLIs), and Error Budgets, achieving high system reliability

DXC Technology

DevOps Engineer

Jan 2019 – Apr 2024

- Designed and implemented Groovy-based CI/CD pipelines to automate software builds, testing, and deployment processes, improving efficiency and reducing manual errors
- Eliminated potential failure points in production architecture by deploying service instances (e.g., EC2 or RDS) across multiple isolated AZs, enhancing system resiliency and availability
- Participated in on-call rotations, resolved incidents promptly, and maintained high availability of AWS-hosted applications, ensuring minimal downtime
- Ensured cloud solutions and deployments aligned with enterprise architecture strategies and standards, supporting consistent business, data, application, infrastructure, network, and security practices
- Reviewed existing architectures to identify technical deficiencies and design gaps, creating actionable recommendations for cloud migration and enhancements, leading to improved system performance
- Managed version control repositories using Git and coordinated code reviews, ensuring a high-quality codebase and reducing integration issues
- Monitored system performance using AWS CloudWatch and Grafana, conducting periodic capacity planning exercises to optimize resource allocation and prevent downtime
- Led the design and implementation of Docker-based containerization strategies for microservices architectures, resulting in improved scalability and simplified deployment processes.
- Automated the deployment and testing of resources using Infrastructure as Code (Terraform and CloudFormation) through pipelines, allowing customers to rapidly build, test, and release code while minimizing errors
- Deployed Istio service mesh to manage microservices communication, enhance security, and streamline traffic routing, leading to improved service reliability and security
- Configured Calico for network policy enforcement, which isolated application workloads and enhanced cluster security
- Implemented Horizontal Pod Autoscaling (HPA) based on CPU and memory metrics, optimizing resource utilization and improving application performance
- Created Helm charts for various applications, which improved deployment consistency and simplified rollbacks
- Utilized Sumo Logic for log aggregation, analyzing trends, and identifying anomalies, which enabled proactive issue resolution
- Created scripts for system administration and AWS using BASH and Python, and developed Lambda functions to upload code and check changes in S3 and DynamoDB, which streamlined processes and improved efficiency
- Enforced regulatory standards (PCI, SOC2, CIST BENCHMARK) using AWS Config, ensuring compliance and enhancing security posture
- Integrated SonarQube for quality testing and Clamav for antivirus protection, improving code quality and system security

Axle Informatics

Solutions Architect

Nov 2016 - Dec 2018

- Architected a staging environment for migration using Terraform, enhancing system readiness and reducing downtime
- Utilized Auto-scaling and load balancers with Terraform to distribute traffic evenly, improving system performance and reliability
- Developed Terraform modules to streamline infrastructure deployment, reducing setup time and minimizing errors
- Onboarded AWS assets using Dome9 for enhanced security visibility, posture management, and IAM management, improving security compliance
- Architected and created three tiers of architecture via Terraform for migrations, facilitating seamless transitions and improving system scalability
- Created IAM for usage across the environment, enhancing security and access management
- Provisioned and deployed computer and networking infrastructure using CloudFormation and Terraform, improving scalability and efficiency
- Architected and maintained a secure environment using IAM, CloudTrail, and Access server login, ensuring data protection and compliance
- Architected and managed storage solutions such as EC2 and EBS volumes, optimizing data storage and retrieval
- Developed baseline and custom guardrails, policies, centralized policy enforcement, tagging policies, and a well-architected multi-account environment, enhancing governance and operational efficiency
- Used AWS Config and Entrust at an enterprise level to enforce MFA across accounts, resources, and emails, enhancing security within our environment
- Leveraged AWS Config to ensure compliance and remediations across multiple accounts within an AWS organization, improving overall system integrity
- Designed and implemented highly available, scalable, and secure AWS infrastructures using IAC tools such as Terraform, resulting in improved system reliability and scalability
- Utilized code repositories (GitHub) and CI/CD tool (Jenkins) to enhance performance on the CI/CD pipeline using the

master-slave configuration, leading to faster deployment times.

- Ensured security on the CI/CD pipeline and EKS cluster by provisioning temporal environments for developers using containerization tools like Docker, which enhanced development efficiency and security

Education

- **National Polytechnic University**
- Bachelor's, computer science

Certifications

- **AWS Certified Solutions Architect - Professional**
- **Professional Scrum Master Certification**
- **Professional Scrum Product Owner Certification**
- **Terraform Certified Associate**

